Advanced Control Systems — Assignment 2
Energy Formulation and Dynamic Modeling of Serial Manipulators

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Abstract

This report details the energy-based dynamic formulation for serial robot manipulators, focusing on the 3-DOF PRR manipulator. We present the analytical definitions of kinetic and potential energy for rigid bodies, review König's theorem and Steiner's parallel axis theorem, and construct the generalized manipulator inertia matrix $\mathbf{B}(\mathbf{q})$. The mathematical formulation is systematically cross-checked against symbolic derivations implemented in MATLAB.

1 Kinetic Energy of Rigid Link Bodies

Under the standard assumptions of rigid links and rigid actuation transmissions, the total kinetic energy $\mathcal{T}$ of an n-degree-of-freedom manipulator is the sum of the kinetic energies of each individual link $\mathcal{T}_{l_i}$:

$$\mathcal{T}(\mathbf{q}, \mathbf{q}) = \sum_{i=1}^{n} \mathcal{T}_{l_i}(\mathbf{q}, \mathbf{q}) \tag{1}$$

For link i with volume V_{l_i} and continuous mass density distribution $\rho(\mathbf{r})$, the kinetic energy is defined as the volume integral of the squared magnitude of the point velocities $\mathbf{p}_i^*$:

$$\mathcal{T}_{l_i} = \frac{1}{2} \int_{V_{l_i}} (\mathbf{p}_i^*)^T \mathbf{p}_i^* \, \rho \, dV \tag{2}$$

Let $\mathbf{p}_{l_i} \in \mathbb{R}^3$ be the position of the center of mass (CoM) of link i with respect to the base frame Σ_0, and let $\mathbf{r}_i = \mathbf{p}_i^* - \mathbf{p}_{l_i}$ denote the relative position vector of any internal point. The velocity field inside the rigid link is governed by:

$$\mathbf{p}_i^* = \mathbf{p}_{l_i} + \boldsymbol{\omega}_i \times \mathbf{r}_i = \mathbf{p}_{l_i} + \mathbf{S}(\boldsymbol{\omega}_i)\mathbf{r}_i \tag{3}$$

where $\boldsymbol{\omega}_i$ is the link angular velocity and $\mathbf{S}(\boldsymbol{\omega}_i)$ is the skew-symmetric cross-product operator.

2 König's Theorem and Kinetic Energy Decomposition

Substituting the velocity field into the kinetic energy integral yields three distinct contributions:

$$\mathcal{T}_{l_i} = \underbrace{\frac{1}{2} \int_{V_{l_i}} \mathbf{p}_{l_i}^T \mathbf{p}_{l_i} \, \rho \, dV}_{\text{Translational Energy}} + \underbrace{\int_{V_{l_i}} \mathbf{p}_{l_i}^T \mathbf{S}(\boldsymbol{\omega}_i)\mathbf{r}_i \, \rho \, dV}_{\text{Mutual Energy}} + \underbrace{\frac{1}{2} \int_{V_{l_i}} \mathbf{r}_i^T \mathbf{S}(\boldsymbol{\omega}_i)^T \mathbf{S}(\boldsymbol{\omega}_i)\mathbf{r}_i \, \rho \, dV}_{\text{Rotational Energy}} \tag{4}$$

2.1 Mutual Energy Cancellation

Since the linear velocity of the CoM $\mathbf{p}_{l_i}$ and the angular velocity $\boldsymbol{\omega}_i$ do not depend on the spatial integration variable dV, the mutual term can be rewritten as:

$$\int_{V_{l_i}} \mathbf{p}_{l_i}^T \mathbf{S}(\boldsymbol{\omega}_i)\mathbf{r}_i \, \rho \, dV = \mathbf{p}_{l_i}^T \mathbf{S}(\boldsymbol{\omega}_i) \left[\int_{V_{l_i}} (\mathbf{p}_i^* - \mathbf{p}_{l_i}) \, \rho \, dV \right] \tag{5}$$

By definition of the center of mass, $\int_{V_{l_i}} \mathbf{p}_i^* \, \rho \, dV = m_{l_i} \mathbf{p}_{l_i}$. Consequently:

$$\int_{V_{l_i}} (\mathbf{p}_i^* - \mathbf{p}_{l_i}) \, \rho \, dV = m_{l_i} \mathbf{p}_{l_i} - m_{l_i} \mathbf{p}_{l_i} = \mathbf{0} \tag{6}$$

The mutual cross-coupling term vanishes identically.

2.2 König's Theorem Statement

König's Theorem states that the total kinetic energy of a rigid body is decoupled into two independent terms:

1. The **translational kinetic energy** of a fictitious point mass concentrated at the center of mass moving at velocity $\mathbf{p}_{l_i}$:
$$\mathcal{T}_{\text{trans}} = \frac{1}{2} m_{l_i} \mathbf{p}_{l_i}^T \mathbf{p}_{l_i} \tag{7}$$
2. The **rotational kinetic energy** associated with body rotation about its center of mass:
$$\mathcal{T}_{\text{rot}} = \frac{1}{2} \boldsymbol{\omega}_i^T \mathbf{I}_{l_i} \boldsymbol{\omega}_i \tag{8}$$

where $\mathbf{I}_{l_i} = \int_{V_{l_i}} \mathbf{S}(\mathbf{r}_i)^T \mathbf{S}(\mathbf{r}_i) \rho \, dV$ is the inertia tensor relative to the CoM expressed in the base reference frame Σ_0.

Because $\mathbf{I}_{l_i}$ in frame Σ_0 is configuration-dependent ($\mathbf{I}_{l_i}(\mathbf{q}) = \mathbf{R}_i(\mathbf{q}) \mathbf{I}_{l_i}^i \mathbf{R}_i^T(\mathbf{q})$), where $\mathbf{I}_{l_i}^i$ is the constant link-frame inertia tensor, König's formulation is compactly written as:

$$\mathcal{T}_{l_i} = \frac{1}{2} m_{l_i} \mathbf{p}_{l_i}^T \mathbf{p}_{l_i} + \frac{1}{2} \boldsymbol{\omega}_i^T \mathbf{R}_i \mathbf{I}_{l_i}^i \mathbf{R}_i^T \boldsymbol{\omega}_i \tag{9}$$

3 Steiner's Theorem (Parallel Axis Theorem)

3.1 Physical Meaning and Statement

Let $\mathbf{I}_C$ denote the inertia tensor of a rigid body of mass m computed with respect to a reference frame Σ_C centered directly at its center of mass. **Steiner's Theorem** (the Parallel Axis Theorem) states that if the coordinate frame is translated by a displacement vector $\mathbf{r} \in \mathbb{R}^3$ without changing orientation to a new frame Σ', the new inertia tensor $\mathbf{I}$ is given by:

$$\mathbf{I} = \mathbf{I}_C + m \, \mathbf{S}^T(\mathbf{r}) \mathbf{S}(\mathbf{r}) = \mathbf{I}_C + m \left((\mathbf{r}^T \mathbf{r}) \mathbf{I}_{3\times 3} - \mathbf{r}\mathbf{r}^T \right) \tag{10}$$

3.2 Utility in Robotics Modeling

In kinematic and dynamic chains, CAD models and standard URDF specifications often supply link inertia tensors either at the joint attachment frame Σ_i or directly at the link CoM Σ_{C_i}. Steiner's theorem enables:

- Converting inertia parameters between the link coordinate origin and the CoM.
- Direct computation of the total effective rotational resistance about the physical joint actuation axes.
- Transforming component-level mass distributions into equivalent lumped models.

4 Potential Energy of Rigid Manipulators

In the absence of elastic/flexible elements, potential energy arises exclusively from gravity. For link i, the gravitational potential energy $\mathcal{U}_{l_i}$ corresponds to the work done against the constant gravity acceleration field $\mathbf{g}_0 = [0, 0, -g]^T \in \mathbb{R}^3$ (expressed in the base frame Σ_0):

$$\mathcal{U}_{l_i} = -\int_{V_{l_i}} \mathbf{g}_0^T \mathbf{p}_i^* \, \rho \, dV = -m_{l_i} \mathbf{g}_0^T \mathbf{p}_{l_i} \tag{11}$$

Summing over all n links:

$$\mathcal{U}(\mathbf{q}) = -\sum_{i=1}^{n} m_{l_i} \mathbf{g}_0^T \mathbf{p}_{l_i}(\mathbf{q}) \tag{12}$$

Because the position of the center of mass $\mathbf{p}_{l_i}$ depends strictly on the joint variables up to link i ($\mathbf{p}_{l_i} = \mathbf{p}_{l_i}(q_1, \ldots, q_i)$), the potential energy is purely a nonlinear function of configuration $\mathbf{q}$ and is velocity-independent.

5 Manipulator Inertia Matrix Formulation

Using the link linear Jacobian $\mathbf{J}_P^{l_i}(\mathbf{q}) = \frac{\partial \mathbf{p}_{l_i}}{\partial \mathbf{q}}$ and angular Jacobian $\mathbf{J}_O^{l_i}(\mathbf{q})$ such that $\mathbf{p}_{l_i} = \mathbf{J}_P^{l_i}(\mathbf{q})\mathbf{q}$ and $\boldsymbol{\omega}_i = \mathbf{J}_O^{l_i}(\mathbf{q})\mathbf{q}$, the kinetic energy of link i is:

$$\mathcal{T}_{l_i} = \frac{1}{2}\mathbf{q}^T \left[m_{l_i} (\mathbf{J}_P^{l_i})^T \mathbf{J}_P^{l_i} + (\mathbf{J}_O^{l_i})^T \mathbf{R}_i \mathbf{I}_{l_i}^i \mathbf{R}_i^T \mathbf{J}_O^{l_i} \right] \mathbf{q} \tag{13}$$

Summing over all links yields the classical quadratic form:

$$\mathcal{T}(\mathbf{q}, \mathbf{q}) = \frac{1}{2}\mathbf{q}^T \mathbf{B}(\mathbf{q}) \mathbf{q} \tag{14}$$

where the generalized **Inertia Matrix** $\mathbf{B}(\mathbf{q}) \in \mathbb{R}^{n \times n}$ is:

$$\mathbf{B}(\mathbf{q}) = \sum_{i=1}^{n} \left(m_{l_i} (\mathbf{J}_P^{l_i}(\mathbf{q}))^T \mathbf{J}_P^{l_i}(\mathbf{q}) + (\mathbf{J}_O^{l_i}(\mathbf{q}))^T \mathbf{R}_i(\mathbf{q}) \mathbf{I}_{l_i}^i \mathbf{R}_i^T(\mathbf{q}) \mathbf{J}_O^{l_i}(\mathbf{q}) \right) \tag{15}$$

Properties of $\mathbf{B}(\mathbf{q})$:

- **Symmetry:** $\mathbf{B}(\mathbf{q}) = \mathbf{B}^T(\mathbf{q})$ for all $\mathbf{q} \in \mathbb{R}^n$.
- **Positive Definiteness:** $\mathbf{B}(\mathbf{q}) > 0$ for all physical configurations, ensuring that $\mathcal{T} > 0$ whenever $\mathbf{q} \neq \mathbf{0}$.
- **Lyapunov Candidate:** The quadratic energy form $\frac{1}{2}\mathbf{q}^T \mathbf{B}(\mathbf{q}) \mathbf{q}$ provides a natural energy-based candidate Lyapunov function for non-linear robot control.

6 MATLAB Model Verification

The dynamic quantities derived from the symbolic MATLAB script confirm full adherence to the theoretical properties:

Table 1: Symbolic consistency evaluation of the kinetic and potential energy models

Analytical Check	**Expression Evaluated**	**Result**
Inertia Matrix Symmetry	simplify($\mathbf{B}(\mathbf{q}) - \mathbf{B}^T(\mathbf{q})$)	$\mathbf{0}_{3\times3}$
Quadratic Form Equivalence	simplify $\left(\mathcal{T} - \frac{1}{2}\mathbf{q}^T\mathbf{B}(\mathbf{q})\mathbf{q}\right)$	0
Total Mass Translation	$\mathbf{B}_{11}(q_1)$ for prismatic axis	$m_1 + m_2 + m_3$
Gravity Vector Alignment	$\mathbf{g}_0 = [0, 0, -g]^T \implies \mathcal{U}_i = m_i g p_{l_i,z}$	Verified

7 Conclusion

The kinetic and potential energy formulations for the 3-DOF PRR serial manipulator have been developed based on rigid-body mechanics. König's theorem provides the translational and rotational split of kinetic energy, while Steiner's theorem formalizes reference-frame inertia shifts. The derived inertia matrix $\mathbf{B}(\mathbf{q})$ satisfies the necessary symmetry and positive definiteness conditions, establishing the foundation for dynamic equation derivation via the Euler-Lagrange equations.